CSC311 Project Proposal

Painting Identification from Survey Responses

Submitted by

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1 Executive Summary

We plan to build a 3-class classifier that predicts the painting (*The Persistence of Memory*, *The Starry Night*, *The Water Lily Pond*) from survey responses. In this proposal we summarize the dataset and specify a reproducible evaluation protocol and model families to compare.

2 Data Exploration

Dataset overview

The training data contains 1686 rows and 562 unique students (identified by `unique_id`). Each student contributes exactly three examples (one per class), yielding perfectly balanced classes (Table 1). The prediction target is the `Painting` column (3 classes).

Table 1: Class balance (training data)

Class (Painting)	# Examples
The Persistence of Memory	562
The Starry Night	562
The Water Lily Pond	562

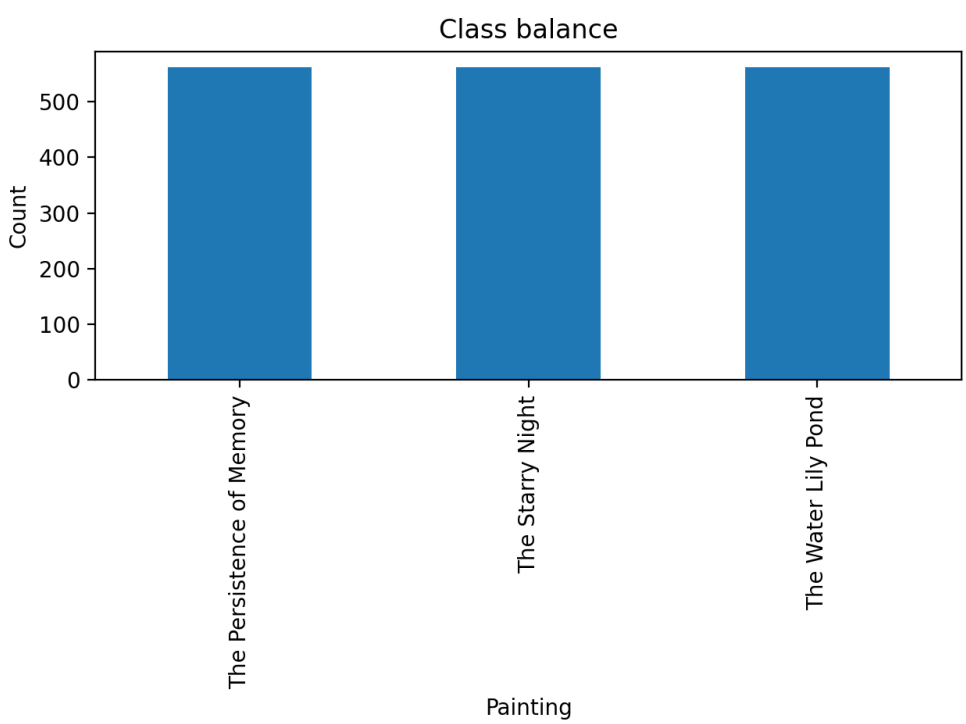

Figure 1: Class distribution across the three paintings (balanced).

Feature types

The dataset includes a mixture of:

- **Numerical**: intensity rating (1–10), counts of prominent colours / objects.
- **Ordinal (Likert)**: four statements (sombre/content/calm/uneasy) encoded as strings such as "4 - Agree".
- **Categorical / multi-select**: room placement, who to view with, season (often stored as comma-separated lists).
- **Free-text**: (i) "Describe how this painting makes you feel" and (ii) "Imagine a soundtrack...".
- **Semi-structured numeric**: willingness-to-pay field containing noisy strings (e.g., "$5", "300 dollars.", etc.).

Missingness and data quality

Missing values are present but modest ($\leq \sim 5\%$ in each feature). Table 2 lists the highest missing-rate columns. We also observed non-standard entries in the payment column (free-form text) and multi-select categorical fields stored as comma-separated strings, requiring careful parsing.

Table 2: Top missing-rate columns (training data)

Column	Missing rate
Imagine a soundtrack for this painting. Describe that soundtrack without naming any objects in the painting.	5.0%
If you could purchase this painting, which room would you put that painting in?	4.7%
If you could view this art in person, who would you want to view it with?	4.5%
What season does this art piece remind you of?	4.5%
Describe how this painting makes you feel.	4.4%
How much (in Canadian dollars) would you be willing to pay for this painting?	4.4%
How many prominent colours do you notice in this painting?	4.4%
How many objects caught your eye in the painting?	4.4%

Text length statistics

The two main text fields are short-to-medium length. For the soundtrack description field, the median length is approximately 8 words (mean 11.6; 95th percentile 34). For the feeling description field, the median length is approximately 8 words (mean 13.7; 95th percentile

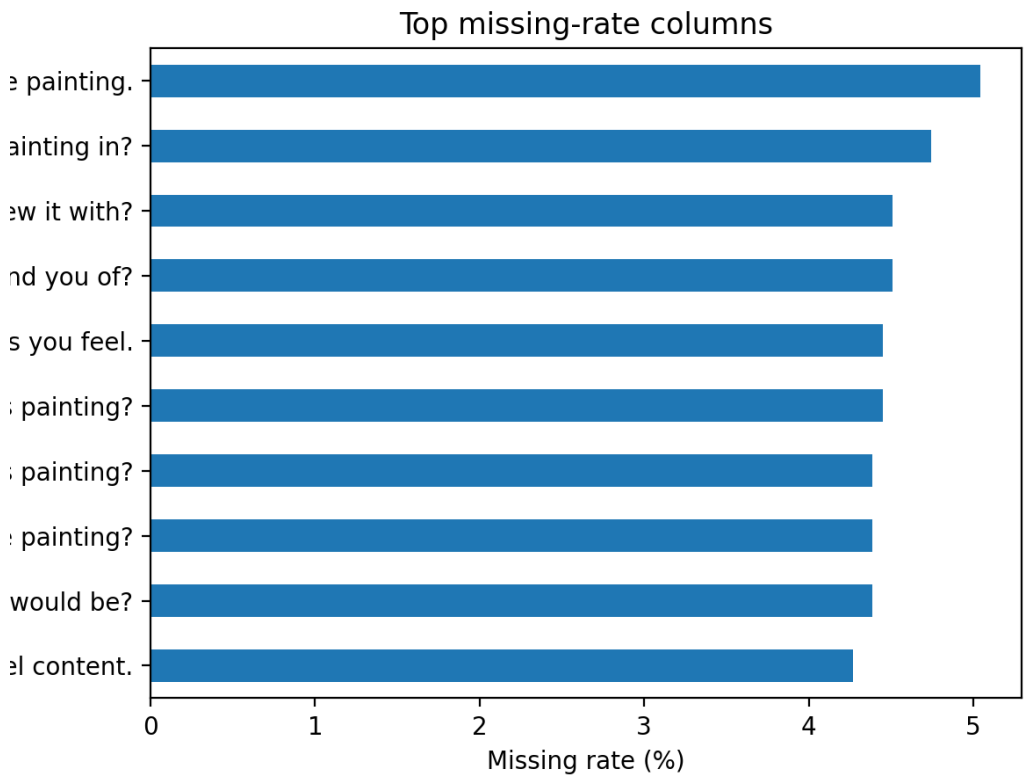

Figure 2: Missing rate by column (top 10).

48). This suggests that bag-of-words / TF-IDF features (possibly with n-grams) should be effective baselines.

Pre-processing plan

We will build a unified preprocessing pipeline:

- **Numerical**: impute missing values using the training-set median; optionally standardize.
- **Likert**: convert strings to integers 1–5 by extracting the leading digit; treat missing as unknown or impute.
- **Payment**: parse the first numeric value from the string via regex; invalid entries become missing and are imputed.
- **Categorical / multi-select**: split comma-separated values and encode using multi-hot vectors with an `UNK` bucket.
- **Text**: lowercase + simple tokenization; vectorize using TF-IDF (word and/or character n-grams).

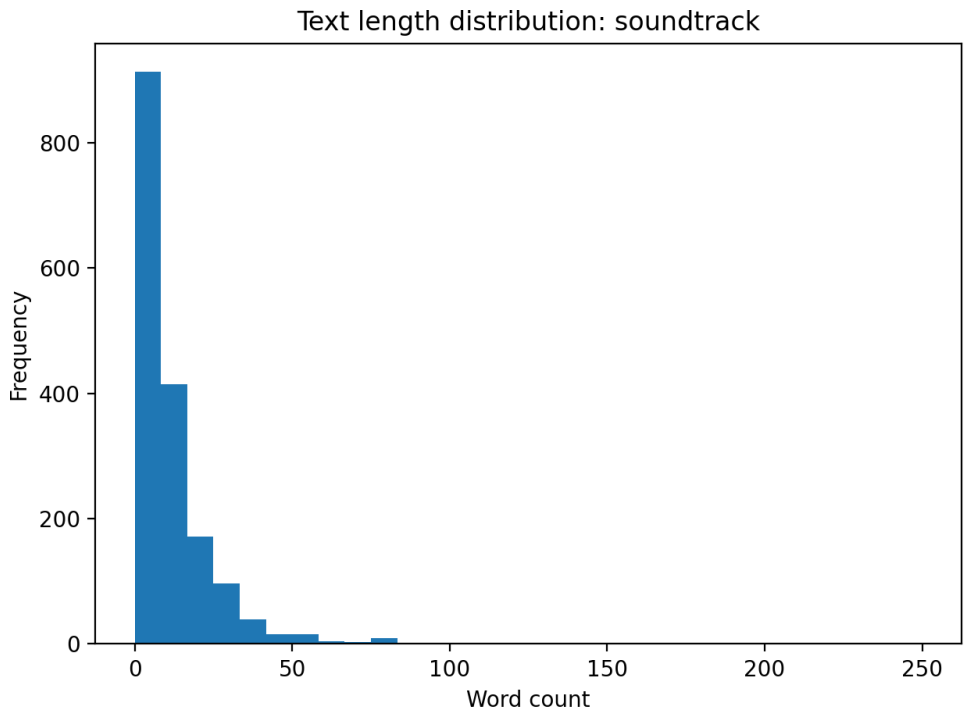

Figure 3: Word-count distribution for the soundtrack description field.

Data splitting and leakage prevention

A key property of this dataset is that each student contributes three closely related examples (one per class). Therefore, we will split **by student** (grouped split): all rows with the same `unique_id` stay together in the same split. We will reserve a held-out test split of students and will not use it during data exploration or hyperparameter tuning.

3 Methodology

Model families

We will compare at least three model families:

1. **Multinomial Naive Bayes (MNB)** on bag-of-words / TF-IDF features (tune smoothing α and n-grams).
2. **Multinomial logistic regression (softmax regression)** on TF-IDF features, optionally concatenated with structured features (tune L2 regularization, vocabulary size, n-grams).
3. **Non-linear baseline: 1-hidden-layer MLP** on the same features (tune hidden size, learning rate, epochs, early stopping).

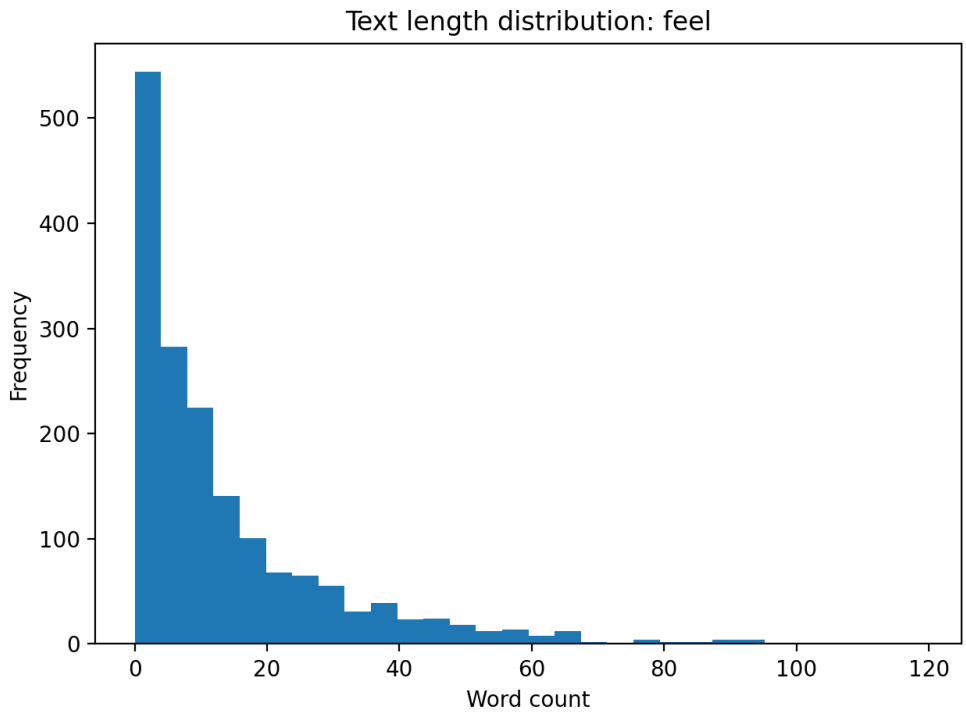

Figure 4: Word-count distribution for the feeling description field.

Optimization, regularization, early stopping

For softmax regression / MLP we will use mini-batch SGD or Adam, L2 regularization, and early stopping based on validation macro-F1.

Validation protocol and hyperparameter tuning

We will use a grouped split by `unique_id`. We will hold out ∼15–20% of students as a final test split (never touched during development). On the remaining students, we will use either a grouped train/validation split or grouped k-fold cross-validation to tune hyperparameters. We will avoid "winner-only tuning" by tuning each family under a comparable budget.

Evaluation metrics

We will report accuracy and at least one additional metric beyond accuracy (e.g., macro-F1) to capture per-class performance.